-: Lab-class Tasks :-

Submitted by :-

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Roll No

36

Program:

BS. CS. 'A'

Submitted To:

Sir. Asad.

Majeed.

Programming Fundamentals Lab -

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TASK - 1.

Run the given programs and obeserve their Outputs.

(a)-

```
# include <iostream>
using namespace std;
int main () {
    char letter;
letter = 'A';
cout << letter << "\n";
letter = 'B';
cout << letter << "\n";
cin >> letter;
return 0;
}.
```

Output:-

A

B.

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(b)-

```
# include <iostream>.
using namespace std;
int main () {
    char letter;
    letter = 65;
  cout << letter << "\n";
   letter = 66;
  cout << letter << "\n";
     cin >> Letter;
     return 0;
}
```

Output:-

A (ASCII code of 65).
B (ASCII code of 66).

(c)-

```
# include <iostream>
using namespace std;
int main ( ) {
  int letter;
  letter = 'A';
cout << letter << "\n";
  letter = 'B';
cout << letter << "\n";
  cin >> letter;
   retur 0;
}
```

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Output:-

65 (ASCII code of A.)
66 (ASCII code of B.)

TASK – 2.

Take two integers as input apply arthematic operations on them (+, -, *, /) as print them on screen.

```
# include < iostream>
using namespace std;
int main ( ) {
    int a = 10;
    int b = 2;
cout << a+b << endl;
cout << a-b << endl;
cout << a * b << endl;
cout << a / b << endl;
}
```

tput:-

12
8
20
5

TASK – 3.

Write the output of the following code

```
#include <iostream>
using namespace std;
int main () {
    int val 1;
    int val 2;
    int val 3;
cout << "Please enter your 3 numbers.";
cin >> val 1 >> val 2 >> val 3;
cout << val 1 << "/n" << val 2 << "\n" <<
                                val 3 << "\n";
cout << val 3 << "/n" << val 2 << "\n" <<
                                val 1 << "\n";
    return 0;
}
```

Output :-

Please enter your 3 numbers are 1, 3, 8.

```
1    8
3    3
8    1
```

TASK – 4.

Write a C++ program to calculate the distance between the two points.

Note:- x_1, y_1, x_2, y_2 are all double values.

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```
# include <iostream>
# include <cMath>
using namespace std;
  int main () {
    double x1, y1, x2, y2 ;
       double distance;
    cout << "Enter x1:";
     cin >> x1;
    cout << "Enter y1:";
     cin >> y1;
    cout << "Enter x2:";
     cin >> x2;
    cout << "Enter y2:";
     cin >> y2;
    distance = sqrt (Pow (x2-x1 ,2) + Pow(y2-y1
                                    ,2));
     cout << "Distance between two points
    is:" << distance << endl;
        return 0;
    }
```

Output:-

```
Enter x1 : 1
Enter y1 : 2
Enter x2 : 4
Enter y2 : 6
```

Distance between two points is 5.

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TASK - 5.

- Calculate the area of circle (area = pie $\times r^2$).

```
# include <iostream>
using namespace std;
int main ( ) {
    double r , area;
    const double pie = 3.14159 ;
    Cout << " Enter radius of circle: " ;
     cin >> r ;
    area = pie * r * r ;
    cout << " Area of a circle : " << area <<
                                        end l;
     return 0;
   }
```

Output:-

Enter radius of circle : 5.
Area of a circle : 78.5397.

- Calculate the area of rectangle (area = l x w

```
# include <iostream>
using namespace std;
int main ( ) {
     float length , width , area;
cout << " Enter length of rectangle : " ;
 cin >> length ;
```

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```
cout << "Enter width of rectangle:";
  cin >> width;
  area = length * width;
  cout << Area of rectangle :" << area << endl;
     return 0;
     }
```

Output:-

```
Enter length of rectangle : 4.2
Enter width of rectangle = 5.1
area of rectangle : 21.42
```

- Calculate the Area of triangle (area = b x h x 0.5)

```
# include <iostream>
using namespace std;
 int main() {
    float base, height, area;
    cout << "Enter base of triangle:";
     cin >> base;
    Cout << "Enter height of triangle:";
      cin >> height;
    area = base x height x 0.5
   Cout << "Area of triangle:" << area <<
                                    endl;
      return 0;
   }
```

Output:-

```
Enter base of triangle : 5.6
Enter height of triangle : 8.1
```

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TASK - 6.

Total purchase A customer in a store is purchasing five items, The Prize of items are.

Prize of item 1: $ 12.95.
Prize of item 2: $ 24.95.
Prize of item 3: $ 6.95.
Prize of item 4: $ 14.95.
Prize of item 5: $ 3.95.

write a program that hold the prize five items in five variables, Display each item prize subtotal of the scale, the amount of scales tax and the total Assumes of scales tax; 6%.

```
# include <iostream>
using namespace std;
 int main () {
    float item 1 = 12.95 f ;
    float item 2 = 24.95 f;
    float item 3 = 6.95 f;
    float item 4 = 14.95 f;
    float item 5 = 3.95 f;
float subtotal , sales tax , total ;
 const float Tax-rate = 0.06 f ;
Subtotal = item 1 + item 2 + item 3 + item 4 +
                                        item 5;
  Sales Tax = subtotal + Tax-Rate;
  Total = Subtotal + Sales Tax;
```

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```
cout << "Price of item 1: $" << item1 << endl;
cout << "Price of item 2: $" << item2 << endl;
cout << "Price of item 3: $" << item3 << endl;
cout << "Price of item 4: $" << item4 << endl;
cout << "Price of item 5: $" << item5 << endl;
cout << "Subtotal of the sale: $" << subtotal <<
cout << "Amount of saleTax (6%): $" << saletax <<
                                                 endl;
cout << "Total: $" << Total << endl;              endl;
    return 0;
}
```

Output:-

```
Price of item 1 : $ 12.95
Price of item 2 : $ 24.95
Price of item 3 : $ 6.95
Price of item 4 : $ 14.95
Price of item 5 : $ 3.95
Subtotal of the sale : 63.75
Amount of Sales Tax : 3.825
Total : $ 67.575
```

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Home – Task:-

1)

- write a program that reads two varriable and swap the value of these varriable.

```
#include <iostream>
using namespace std;
int main() {
    int a, b, temp;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    temp = a;
    a = b;
    b = temp;
    cout << After swaping: " << end;
    cout << " a = " << a << end;
    cout << " b = " << b << end;
    return 0;
}
```

output:

```
Enter two numbers: 10, 20
After swaping:
a = 20
b = 10
```

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•) write a program that prompts the user to enter an alphabet in small caps and display the entered alphabet into upper caps.

```
#include <iostream>.
#include <cctype>
using namespace std;
int main ()
  {
    char alphabet;
cout << "Enter a lower case alphabet:";
cin >> alphabet;
cout << " uppercase alphabet:";
     << char(toupper(alphabet))<<endl;
  return 0;
```

output:

```
Enter lowercase alphabet: a
uppercase : A.
```

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-: Bonus- Task:-

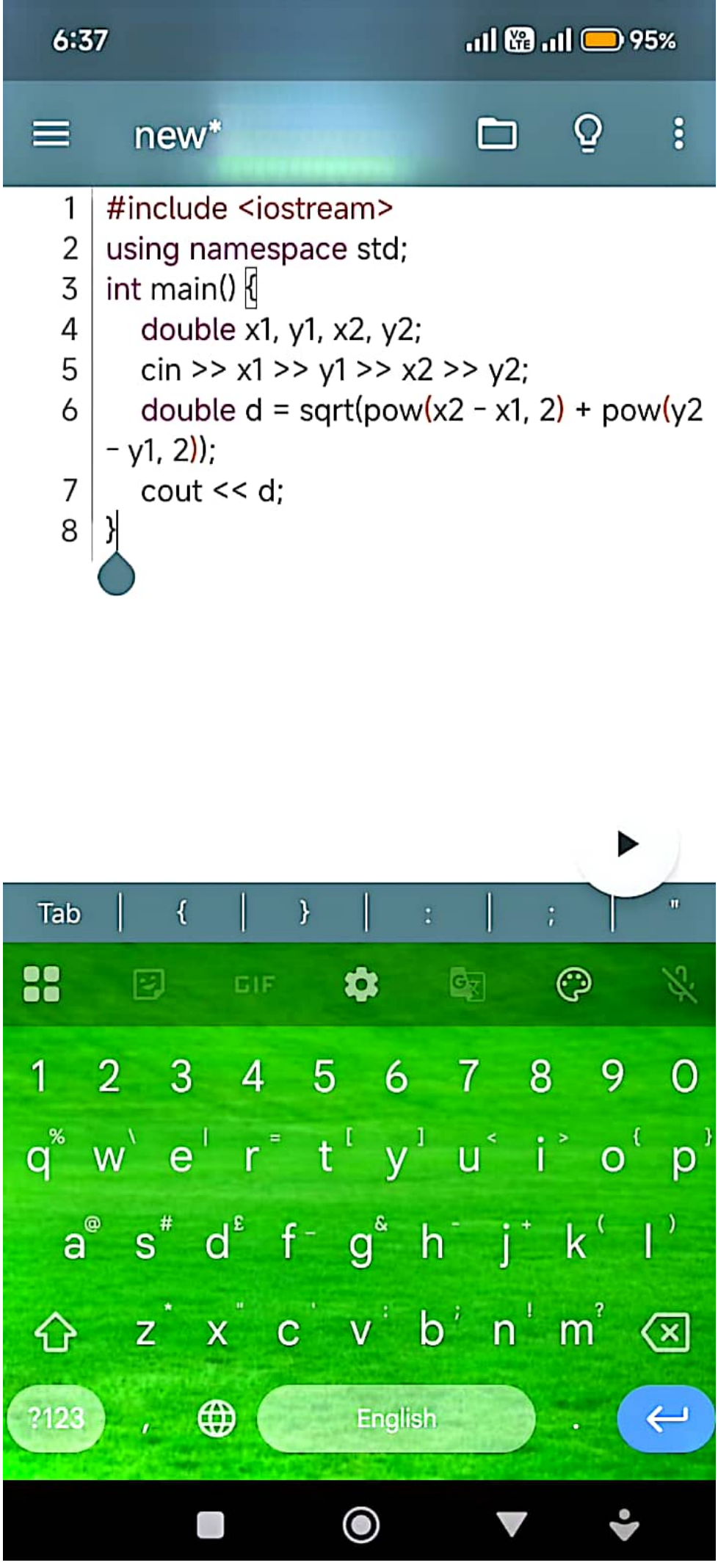
1) write a program that input five-digits integers, separate the integer into its individuals digits, and prints the digits separate from one another by three-spaces.

```
#include<iostream>
using namespace std;
int main() {
    int number;
cout<<"Enter a five-digit integer:";
 Cin>> number;
Cout<< number/10000<<"  "
 << (number/1000)%10<<"  "
 << (number/100)%10<<"  "
<<(number%10<<end;
return 0;
}
```

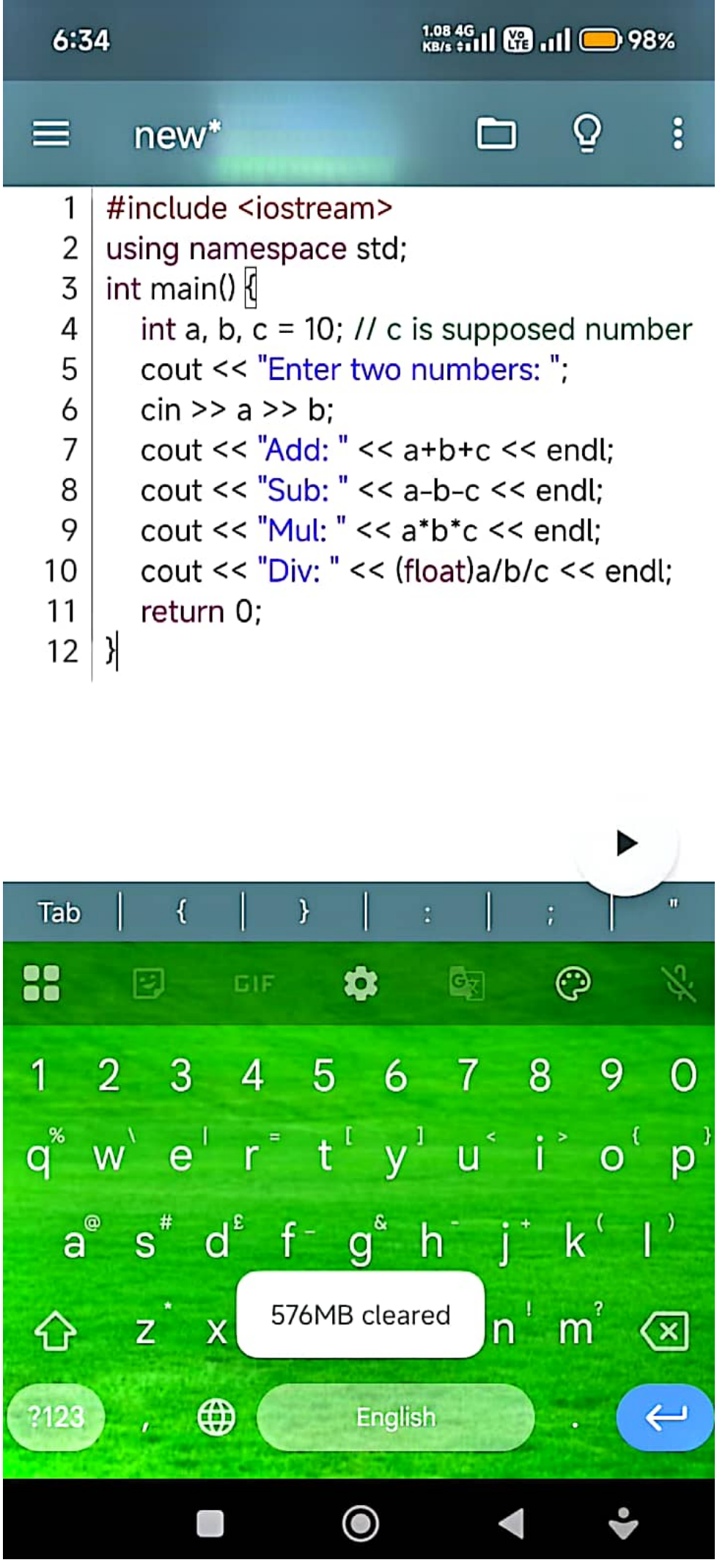
Output:-

```
Input number: 42317
            : 4 2 3 1 7
```

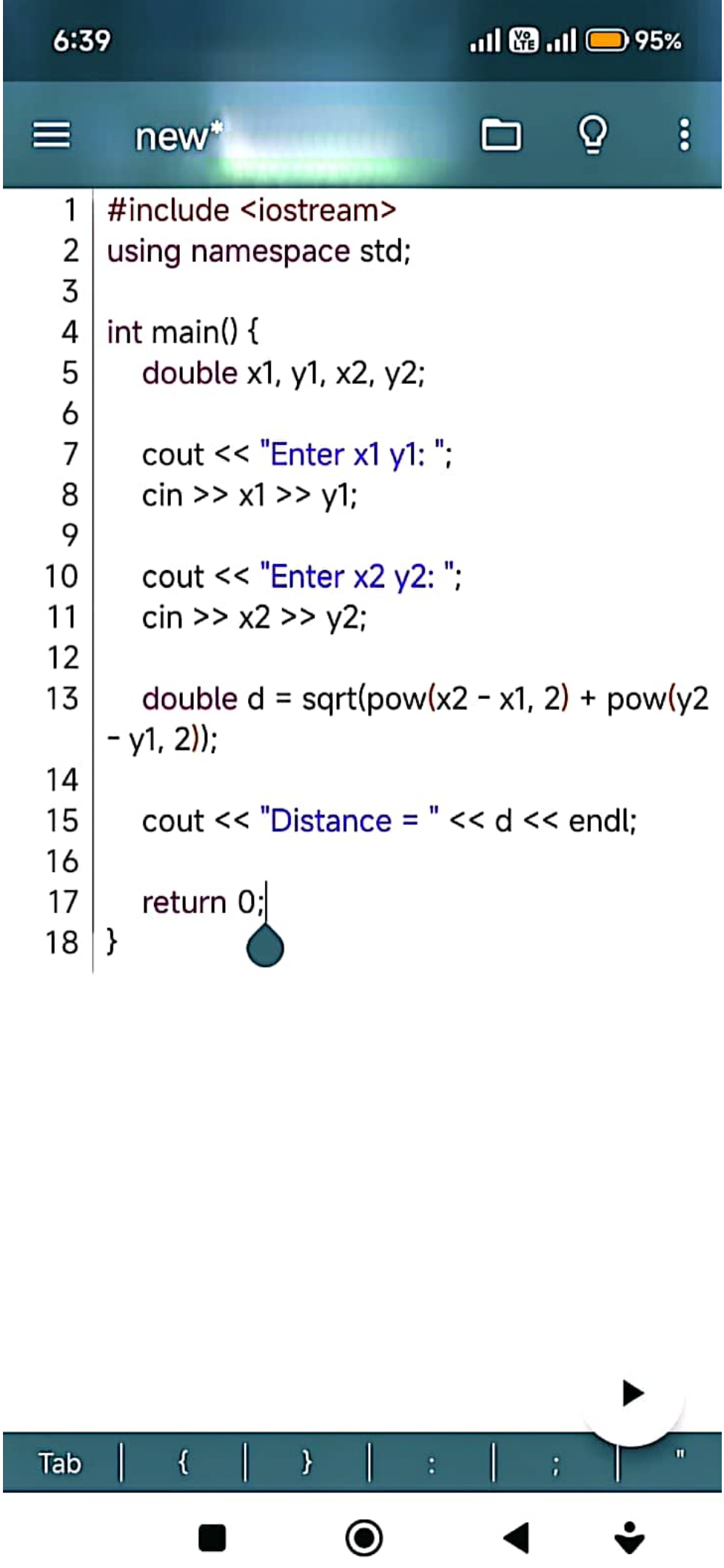
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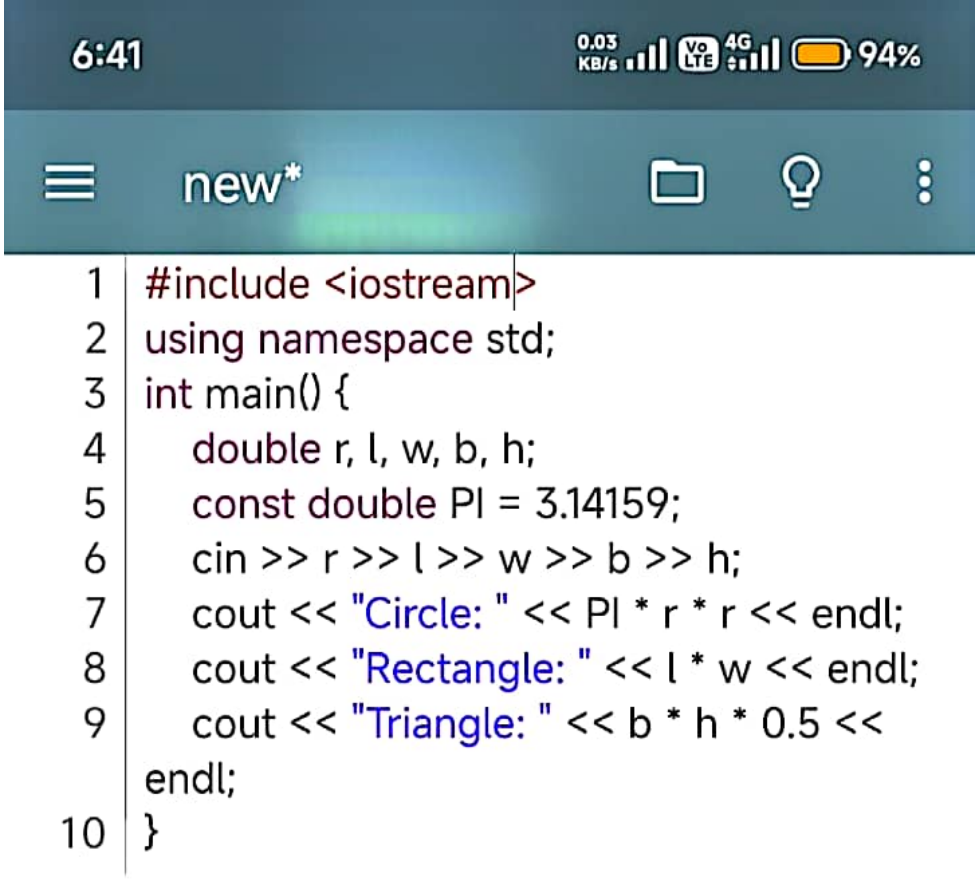
Scanned with CamScanner



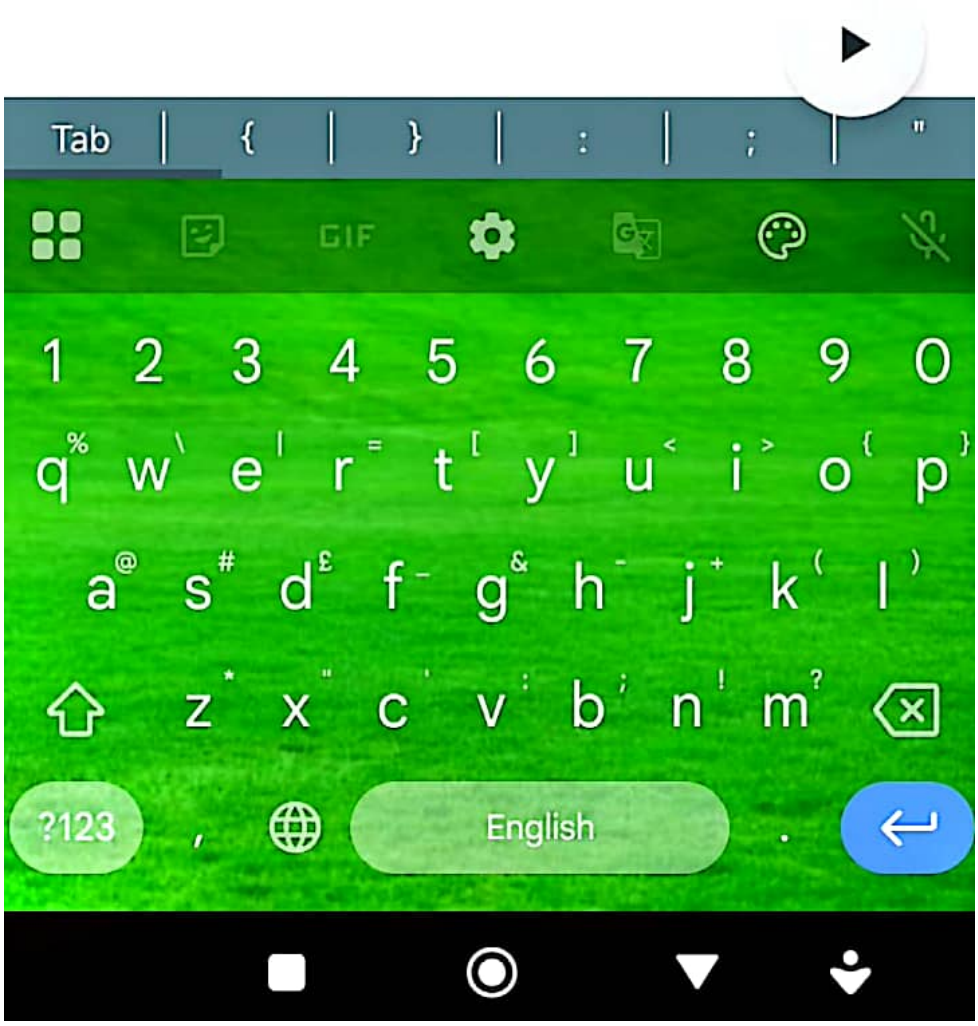
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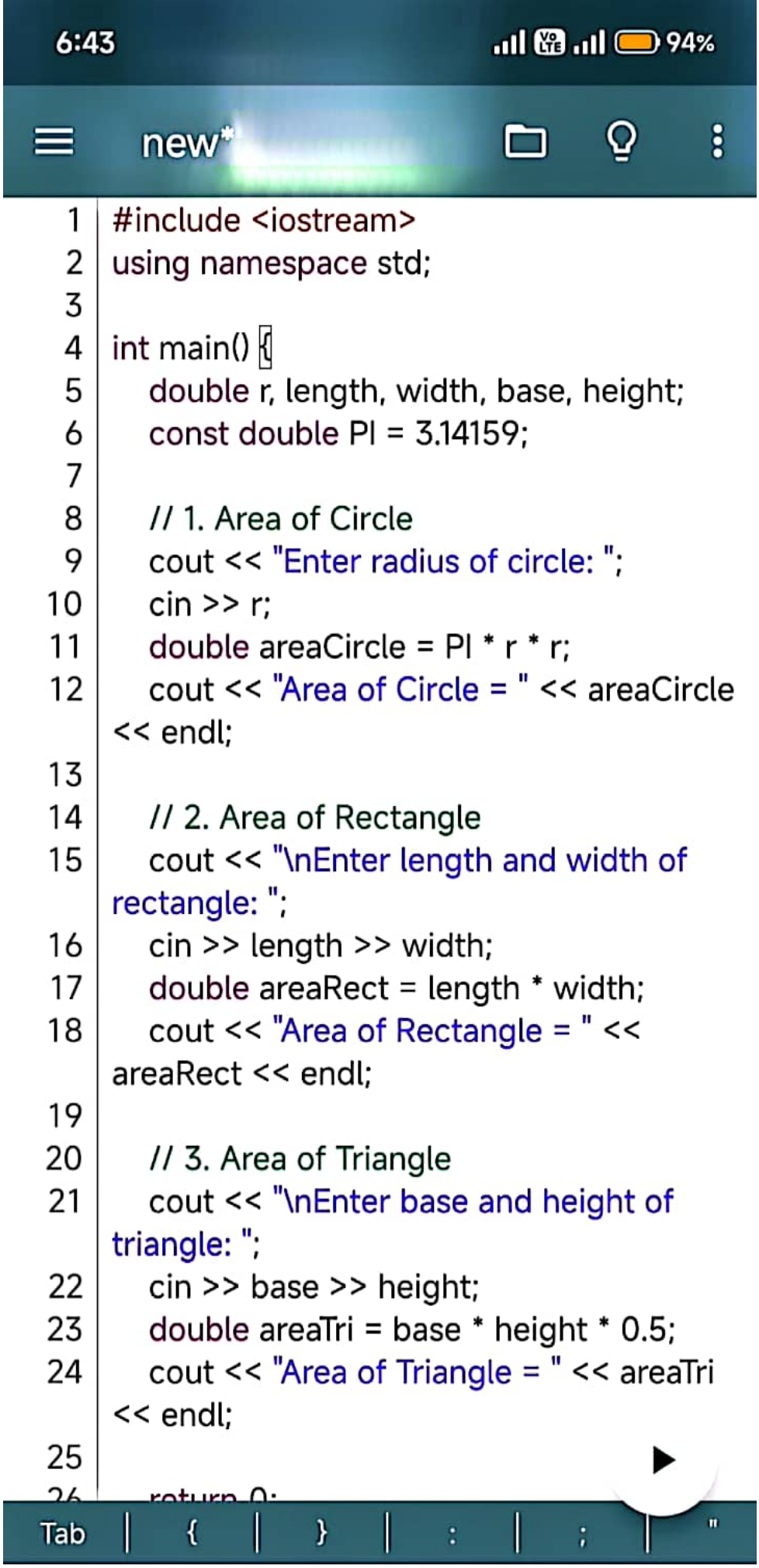
Scanned with CamScanner



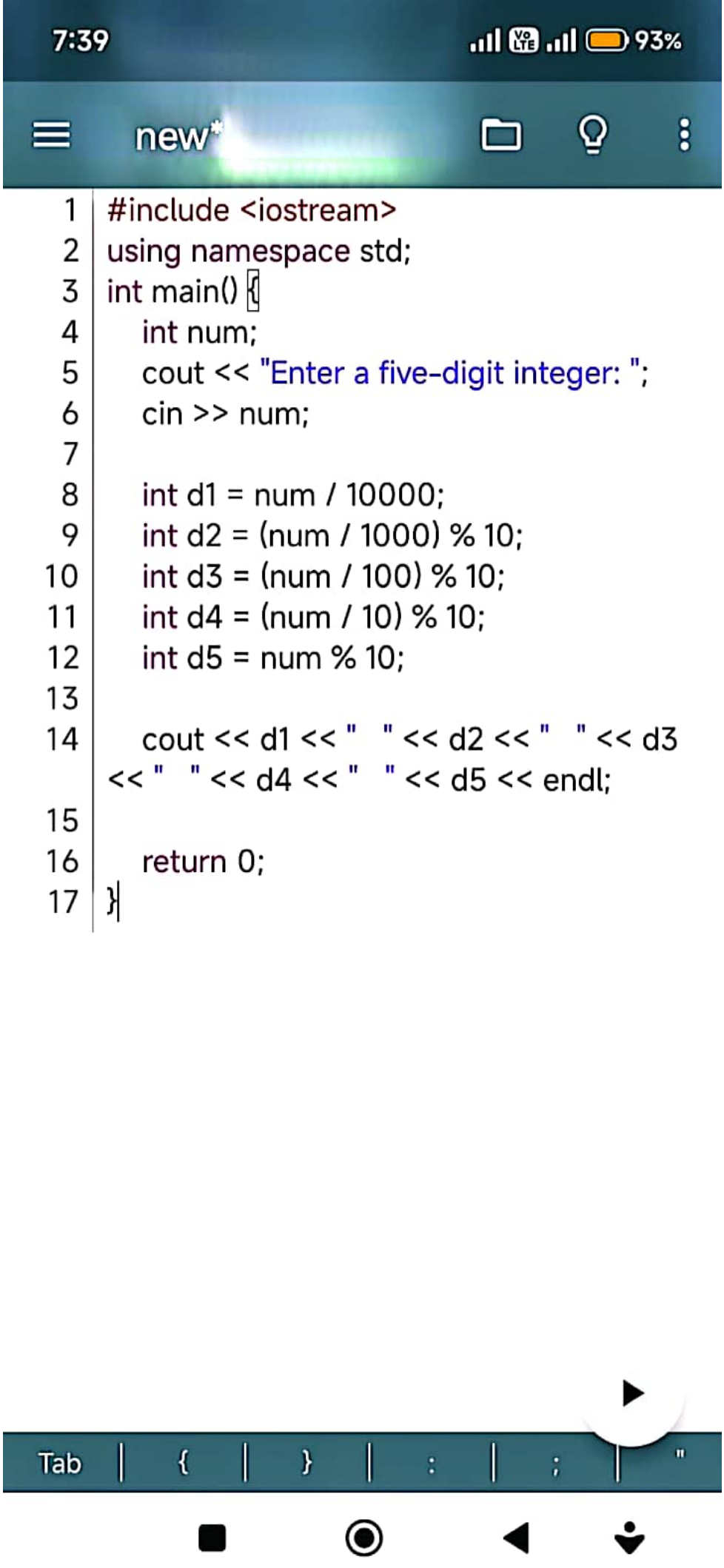

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      double r, l, w, b, h;
5      const double PI = 3.14159;
6      cin >> r >> l >> w >> b >> h;
7      cout << "Circle: " << PI * r * r << endl;
8      cout << "Rectangle: " << l * w << endl;
9      cout << "Triangle: " << b * h * 0.5 <<
   endl;
10 }
```



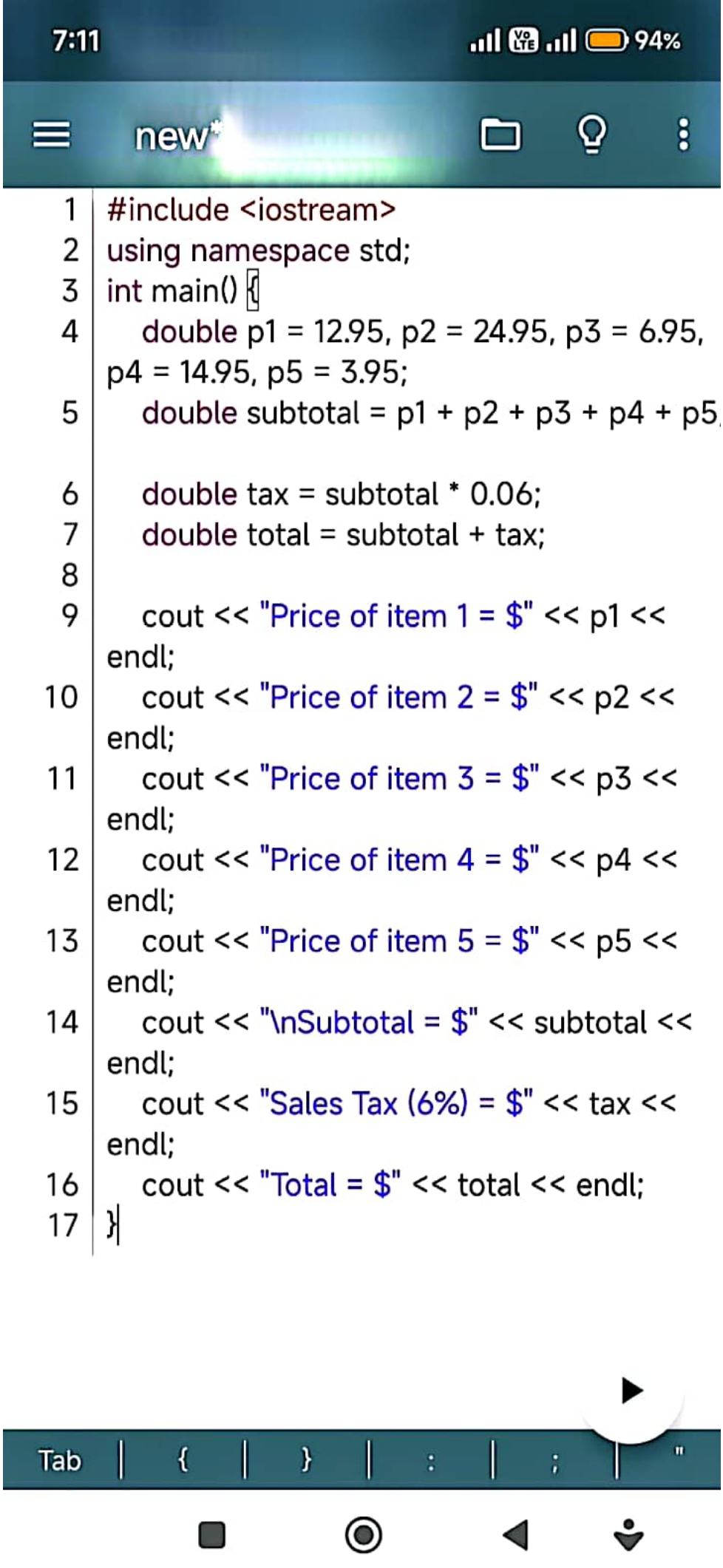
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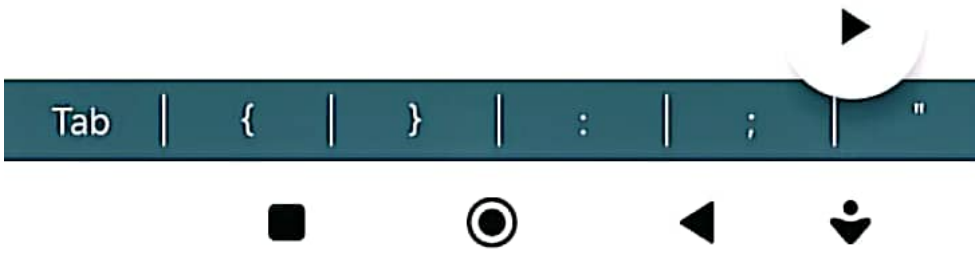


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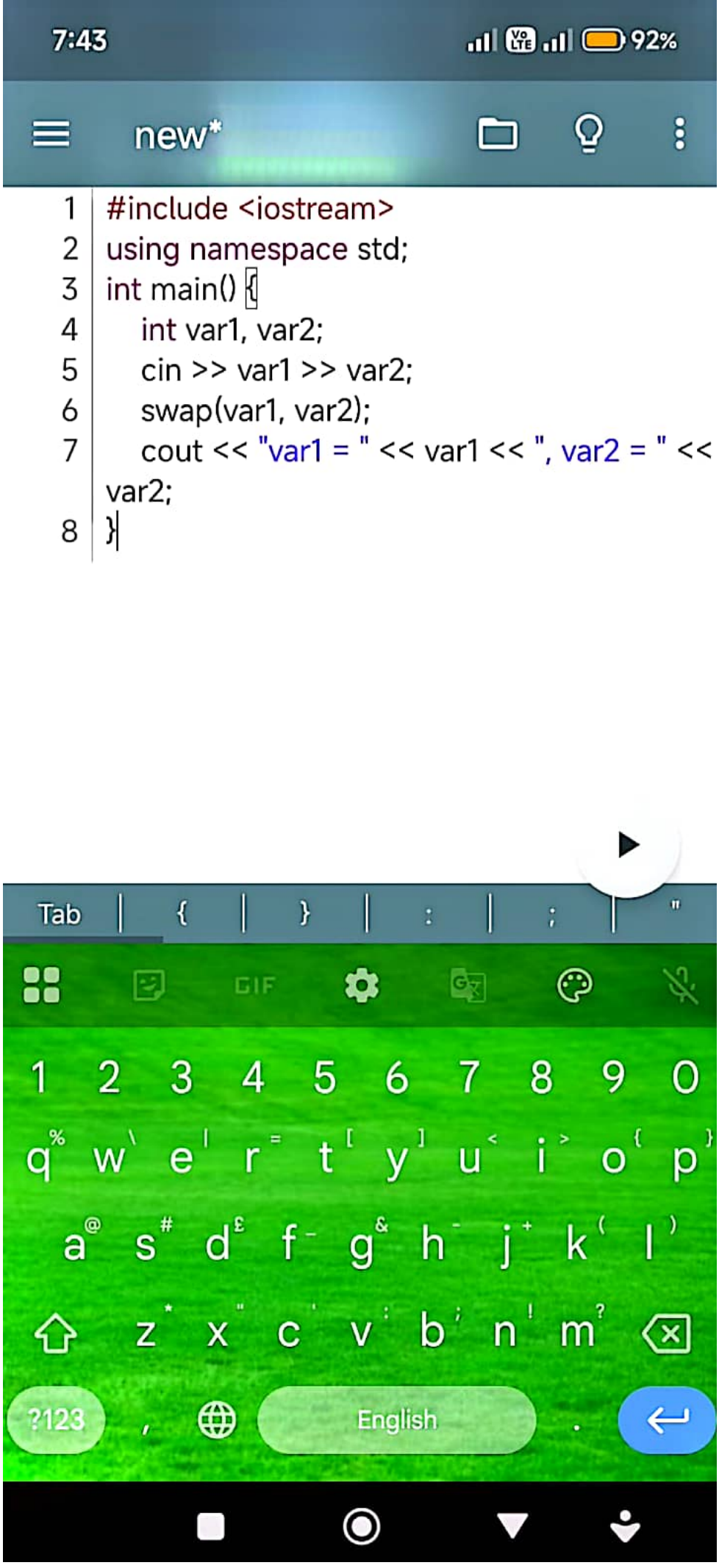



```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      double r, l, w, b, h;
5      double PI = 3.14159;
6
7      cout << "Enter radius of circle: ";
8      cin >> r;
9      cout << "Area of Circle = " << PI * r * r
   << endl;
10
11     cout << "\nEnter length and width of
   rectangle: ";
12     cin >> l >> w;
13     cout << "Area of Rectangle = " << l * w
   << endl;
14
15     cout << "\nEnter base and height of
   triangle: ";
16     cin >> b >> h;
17     cout << "Area of Triangle = " << b * h * 0.
   5 << endl;
18 }
```

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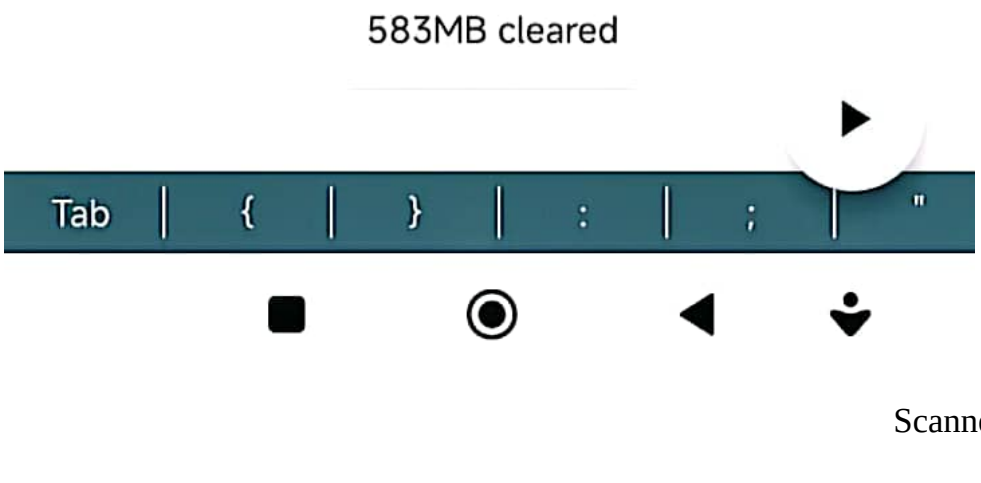



```cpp
#include <iostream>
using namespace std;
int main() {
    int var1, var2;
    cout << "Enter var1: ";
    cin >> var1;
    cout << "Enter var2: ";
    cin >> var2;

    cout << "\nBefore Swap: var1 = " << var1
<< ", var2 = " << var2 << endl;

    // Swapping logic
    int temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After Swap: var1 = " << var1 << ",
 var2 = " << var2 << endl;
    return 0;
}
```



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```cpp
#include <iostream>
using namespace std;
int main() {
    int number;

    cout << "Enter a five-digit integer: ";
    cin >> number;

    cout << number / 10000 << "   "
        << (number / 1000) % 10 << "   "
        << (number / 100) % 10 << "   "
        << (number / 10) % 10 << "   "
        << number % 10 << endl;

    return 0;
}
```

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```
Enter a five-digit integer: 42317
4  2  3  1  7

[Program finished]
```

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```cpp

#include <iostream>
using namespace std;

int main() {
    int a, b, temp;

    cout << "Enter two numbers: ";
    cin >> a >> b;

    temp = a;
    a = b;
    b = temp;

    cout << "After swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;

    return 0;
}
```

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```
Enter two numbers: 50 43
After swapping:
a = 43
b = 50

[Program finished]
```

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```cpp
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char alphabet;

    cout << "Enter a lowercase alphabet: ";
    cin >> alphabet;

    cout << "Uppercase alphabet: "
        << char(toupper(alphabet)) << endl;

    return 0;
}

```



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```
Enter a lowercase alphabet: s
Uppercase alphabet: S

[Program finished]
```

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